

Features

- Demonstrates the [LPF Component](#).
- Simple example code to cycle through all operational configurations and modes.

General description

The LPF_Demo demonstrates the operation of the [LPF Component](#). It has API calls to change the LPF configuration at run-time. This includes the 4 corner frequency settings and BYPASS mode.

LPF_Demo Requirements

- PsoC Creator 4.2 or higher.

Device Families Supported

At this time, the [LPF Component](#) is only supported for the PsoC5 family.

Target Boards

The LPF_Demo project was built for the CY8CKIT-059 board. The default pin assignments are:

- Pin 2[2] is used for the Kit user switch to switch LPF modes in the demo.
- Pin 2[4] is the analog output of the first LPF stage.
- Pin 2[5] is the analog output of the second LPF stage.
- Pin 2[6] is the clock output driving the first LPF stage.
- Pin 12[7] is used as the Tx output to the KitProg for the debugging information.

Other PsoC5 boards can be used however you may have to select a different PsoC5 device and you may have to assign other port pins.

Demo Project Functions

The LPF_Demo is a very simple example program to demonstrate the [LPF Component](#) modes of operation and the code involved.

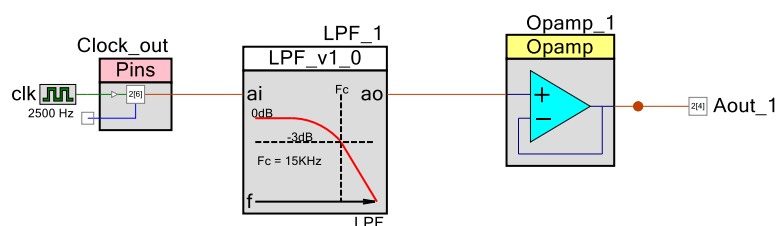
The corner frequencies supported are:

Table 1 - Corner Frequency and RC values

Fc enum value	'R' value	'C' value	Corner Frequency (Fc)
3	1040K Ω	10pF	153 KHz
2	1040K Ω	5pF	77 KHz
1	208K Ω	10pF	31 KHz
0	208K Ω	5pF	15 KHz

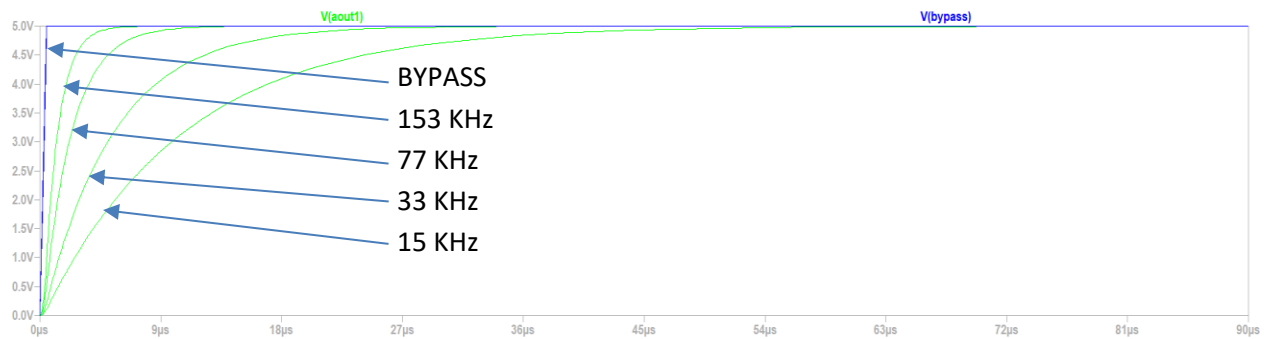
The [LPF Component](#) can be stopped or disabled. When this occurs, the LPF is placed in BYPASS. This mode has no practical analog filtering. The entire frequency spectrum is passed from the input to the output.

Although the [LPF Component](#) to aid in analog signal processing, This project uses it to demonstrate how a low-pass filter can be used to slew-rate control a digital-based signal. In this project I use an internal 2500 Hz clock as the source signal and convert it to an analog signal. I then pass this analog version through the [LPF Component](#). I then amplify this signal through a Opamp in Follower mode.

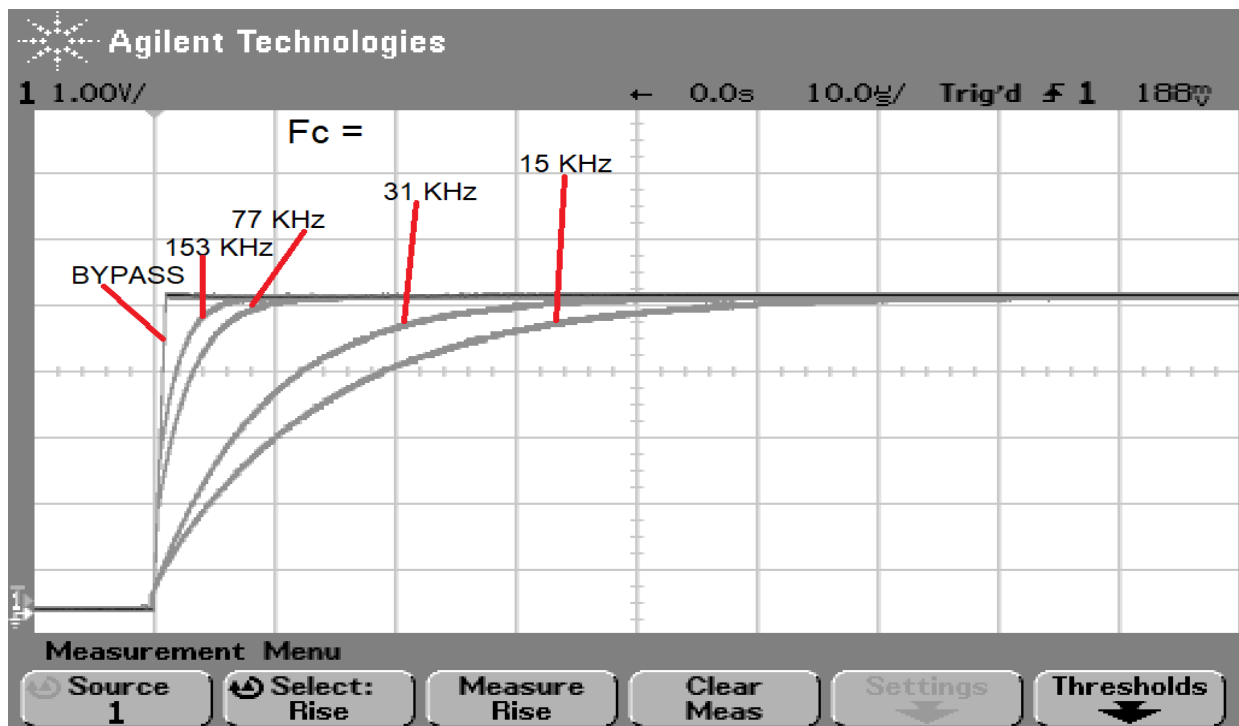


The result is a slew-rate control of both the rise and fall edges of the clock signal by selecting the corner frequency. Using an oscilloscope, it is easy to demonstrate and confirm the operation of the [LPF Component](#).

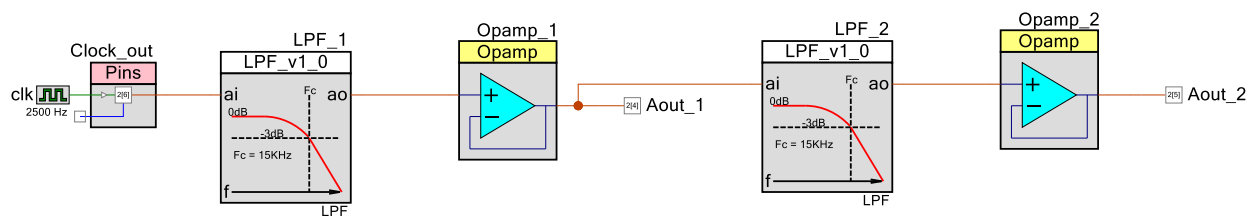
Here are the results of a PSpice simulation of the LPF effect on rise-time slew-rates with the 4 selectable corner frequencies and BYPASS modes. (The same results occur on the fall time slew-rates.)

Figure 1 - PSpice simulation of Fc and BYPASS rise time

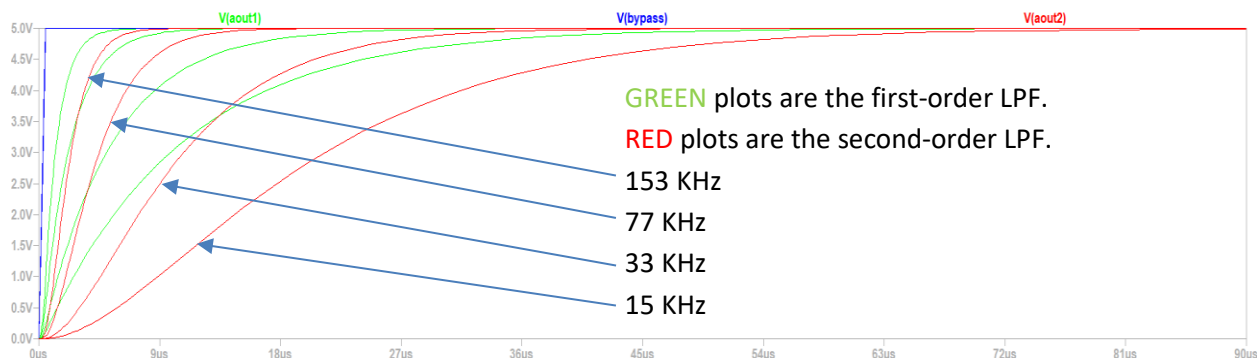
Here are the measured results:

Figure 2 - Measured results of Fc and BYPASS rise time

I've included a second LPF connected to the output of the first LPF to create a second-order LPF implementation.



Here are the results of a PSpice simulation of the second-order LPF effect on rise-time slew-rates with the 4 selectable corner frequencies and BYPASS modes. (The same results occur on the fall time slew-rates.) The serial result of the second-order LPF will further lower the corner frequency.



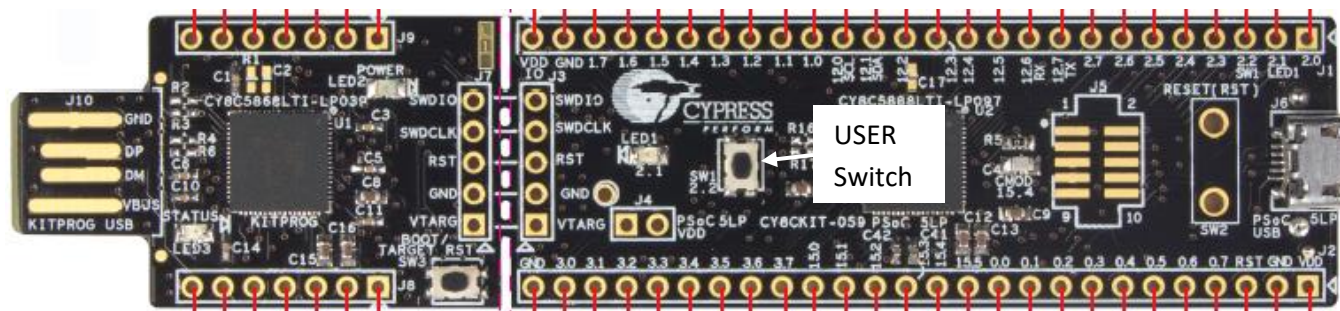
Demo Results

Once the PsoC5 is successfully programmed, connect to it to a terminal program on the Host computer using the assigned COM port. (The connection spec is 115.2Kbps 8N1.)

The LPF_Demo program will print the following information to the terminal window:

```
LPF_Demo 1.0.1
Press the user switch to change the corner frequency...
LPF_1 Fc = 15 KHz   LPF_2 Fc = 15 KHz
```

Press the USER Switch on the CY8CKIT-059 to select the next corner frequency or BYPASS.



```
LPF_1 Fc = 31 KHz   LPF_2 Fc = 31 KHz
LPF_1 Fc = 77 KHz   LPF_2 Fc = 77 KHz
LPF_1 Fc = 153 KHz  LPF_2 Fc = 153 KHz
```

LPF_1 Fc = BYPASS LPF_2 Fc = BYPASS

Project Changes

Version	Description of changes	Reason for changes/impact
1.0.1	first release of the project	

References

LPF Component: LPF_v1_0.pdf	
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